

Brian E. Kane

Full-Stack AI Engineer | Senior Software Engineer | Healthcare Technology

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Professional Summary

Full-stack software engineer with 11 years of enterprise software experience and a current focus on applied AI systems, LLM applications, RAG, agentic workflows, and AI-enabled product development. Former healthcare practitioner and clinical neurology practice leader with a multi-decade clinical background in patient care, healthcare operations, clinical reasoning, and multidisciplinary collaboration. Experienced across React, Java/Spring Boot, Python, FastAPI, AWS, Docker, Terraform, CI/CD, accessibility, cloud migration, and production software delivery. Brings a rare combination of healthcare domain depth, enterprise engineering discipline, applied AI curiosity, mentoring experience, and practical product judgment.

Core Skills

Applied AI: LLM applications, RAG, embeddings, vector databases, agentic workflows, MCP tooling, prompt engineering, context engineering, AI-assisted development

Frontend: React, Next.js, TypeScript, JavaScript, React Router, Redux, Tailwind CSS, responsive UI, accessibility

Backend: Java, Spring Boot, REST APIs, Python, FastAPI, Pydantic, OAuth2, Maven, JUnit, Hibernate

Cloud / DevOps: AWS, Docker, Terraform, CI/CD, Vercel, Render, Pivotal Cloud Foundry to AWS migration

Leadership: Mentoring, stakeholder collaboration, architecture contribution, cross-functional delivery, documentation, code review

Selected AI & Full-Stack Projects

The Hallucination Tribunal

RAG Document Assistant | Multi-Agent Verification Workflow

- Built a RAG-based document assistant that answers questions from a controlled corpus and subjects generated answers to adversarial review.
- Designed a Witness, Prosecutor, and Judge workflow to separate answer generation, claim challenge, and final verdict.
- Demonstrates hybrid retrieval, multi-agent orchestration, claim-level verification, citation-aware responses, document ingestion, and evaluation-focused AI behavior.
- **Tech:** Next.js, TypeScript, Tailwind CSS, Python, FastAPI, Pydantic, ChromaDB, hybrid retrieval, OpenAI, optional Ollama, Vercel, Render.

MediNotes Pro

Healthcare-Focused AI Documentation Assistant

- Built a healthcare-oriented AI assistant that transforms clinician consultation notes into structured visit summaries, doctor next steps, and patient-friendly communication.
- Designed around a PHI/PII-conscious demo architecture using synthetic data and private containerized inference rather than sending notes to third-party LLM APIs.
- Connects practical AI engineering with healthcare workflow knowledge from prior clinical experience.
- **Tech:** Next.js, React, TypeScript, FastAPI, Python, Docker, Ollama, Clerk, SSE streaming, Vercel, Render.

AI Digital Twin

Portfolio AI Assistant | Full-Stack AI Application

- Built a conversational AI assistant grounded in professional background, resume content, project summaries, and curated context.
- Designed to demonstrate practical full-stack AI architecture rather than a generic chatbot demo.
- Uses a FastAPI backend, cloud-hosted model inference, conversation memory, and a browser-based frontend experience.
- **Tech:** Next.js, React, TypeScript, Python, FastAPI, AWS Bedrock, Amazon Nova, Terraform, S3-backed memory, GitHub Actions.

Professional Experience

Express Scripts / The Cigna Group / Evernorth | October 2015 - May 2026

Software Advisor | Senior Software Analyst | Software Analyst | Intern

- Selected as one of eight engineers chosen to help lead the initial modernization of the Express Scripts consumer-facing web platform from a legacy experience into a React and Java/Spring-based application architecture.
- Built and enhanced customer-facing functionality for Order Status, Automatic Refill, Delivery Date Changes, and Over-the-Counter Medications across React front-end applications and Java/Spring Boot backend services.
- Helped improve page load performance by approximately 20%, reduce errors by 25%, increase digital usage by 30-40%, and contribute to a 30% reduction in phone-based orders.
- Worked across front-end, backend, API, and integration layers using React, React Router, Redux, JavaScript, TypeScript, Java, Spring Boot, REST APIs, Maven, JUnit, OAuth2, and Hibernate.
- Contributed to application and infrastructure migration from Pivotal Cloud Foundry to AWS, strengthening cloud deployment, scalability, and production operations experience.
- Helped establish front-end component patterns, implementation standards, and accessible user experiences meeting A11y testing requirements.
- Trained and mentored 10-15 junior and mid-level engineers in React and Java-based full-stack development.
- Partnered daily with business stakeholders, translated requirements into production-ready software, and contributed to architecture decisions within a large-scale enterprise healthcare environment.

Applied AI & Agentic Engineering

- Built hands-on AI workflows using OpenAI models, LLM orchestration, retrieval-augmented generation, vector databases, MCP-based tooling, prompt engineering, context engineering, and autonomous task workflows.
- Developed practical experience with tool-using agents, multi-step orchestration, retrieval pipelines, evaluation-oriented iteration, and deployment-minded AI patterns through recent applied AI projects.
- Used Python-centered AI engineering practices with Dockerized environments, API integration, workflow automation, and full-stack deployment patterns.
- Focused on building AI systems that are useful, explainable, maintainable, and more reliable than generic chatbot demos.

Prior Healthcare Leadership

Clinical Practice / Chiropractic Neurology | 1984 - 2014

Chiropractic Physician | Clinical Neurology Practice Leader | Solo Practitioner

- Practiced in healthcare across solo practice and a large multidisciplinary clinical environment.
- Served in a clinical neurology leadership role, collaborating across disciplines and working with complex patient-care workflows.
- Built deep experience in clinical reasoning, patient communication, documentation, healthcare operations, and interdisciplinary collaboration.
- Developed early hands-on technical experience by networking clinic systems and supporting operational technology needs before transitioning fully into software engineering.
- Brings healthcare workflow understanding and domain credibility to AI, clinical documentation, patient-facing systems, insurance, pharmacy, and healthtech software.

Education

Diplomate, American Board of Chiropractic Neurology - American College of Chiropractic Neurology / Westchester College; Doctor of Chiropractic - Logan University; B.S., Human Anatomy - Logan University; B.S., Human Biology - Logan University; B.S., Zoology, minors in Physics and Chemistry - Eastern Illinois University; additional graduate study in recombinant genetics.

Recent AI Professional Development

The Complete Agentic AI Engineering Course; AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents; AI Builder with n8n: Create Agents & Voice Agents; AI Leader: Generative AI & Agentic AI for Leaders & Founders; AI Engineer Production Track: Deploy LLMs & Agents at Scale; AI Coder: Vibe Coder to Agentic Engineer in 3 Weeks.